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REGULAR PAPER

A visual question answering model based on image captioning

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Abstract
Image captioning and visual question answering are two important tasks in the field of artificial intelligence, which have been widely used in various aspects of life and greatly facilitate our daily life. Image captioning and visual question answering have many similarities and use basically the same related knowledge and techniques. They are both cross-modal tasks involving computer vision and natural language processing, and can be studied in the same model and use the image captioning results to enhance the visual question answering output. Current research on these two tasks has largely been conducted independently, and the accuracy of the visual question answering results needs to be improved. Therefore, this paper proposes a visual question answering model IC-VQA based on image captioning. This model first performs the image captioning part, i.e., obtaining rich visual information by constructing object geometric relations and utilizing mesh information, and then generates question-specific image captioning by means of Attention+ Transformer. Transformer to generate question-specific image captioning sentences. Then the visual question answering part is performed, i.e., the previously generated image captioning sentences are fused to answer the question through the Attention+ LSTM framework, which significantly improves the accuracy of the answer. Experiments on the datasets VQA1.0 and VQA2.0 resulted in an overall accuracy of 70.1 and 70.85, respectively, which significantly closes the gap with humans, which proves the effectiveness of the IC-VQA model, and the accuracy of the visual question answering output can be truly improved by fusing the captioning sentences about the question.

Keywords Image captioning · Visual question answering · Transformer · LSTM

1 Introduction

Deep learning-based image tasks have continuously evolved and advanced, progressing from initial single tasks (such as image segmentation, target detection, etc.) to cross-modal multi-tasks (such as image captioning, visual question answering, etc.). Image captioning and visual question answering, which are cross-applications of computer vision and natural language processing, are current research hotspots, and are now not only interesting research directions, but also have a wide range of applications in the fields of healthcare [1], transportation [2], education [3], and industry [4].

The problem solved by image captioning is to input a picture to the model, and the model automatically generates a text sentence that describes the picture. Image captioning not only needs to recognize the information of the object's type, size, scene, absolute position, and relative positional relationship between the objects, but also generates a natural language sentence that conforms to the content of the image as well as the human habit. sentences, which includes word choice, grammatical structure, contextual relationships, etc.

The main task of visual question answering is to allow computers to understand image content and natural language questions and generate correct answers. Visual question answering requires models with image understanding, natural language processing, multimodal fusion, and reasoning that can simultaneously process the input image and the question, and localize to the relevant object in the image based on the question. Because of the variable form of the problem content, the modeling requirements are more stringent.

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It can be observed that the techniques required for both image captioning and visual question answering tasks are essentially the same, both involving computer vision and natural language processing, and both tasks operate on images and text. Currently, research on these two tasks has largely been conducted independently without intermingling. Assuming that the input to the visual question answering task includes the results of the question and answer, i.e., a sentence containing both the question and the result as a complementary condition, then the results of the question will have an additional source and the accuracy will be improved. So here we are investigating the fusion of these two tasks by generating a descriptive sentence about the question for a particular image and then incorporating this descriptive sentence in the visual question answering to improve the accuracy of the question answering. Figure 1 shows a sample of visual question answering based on image description, where the input is an image and a related question, and the output is a descriptive sentence for the question and the corresponding answer.

The contributions of this paper are as follows:

- We propose a relatively challenging new task that combines image captioning and visual question answering to improve the accuracy of visual question answering through image-specific descriptive sentences.
- A question-guided generation of image captioning module is designed, which can be used to generate image captioning sentences about a specific question through an attention mechanism by constructing object geometric relations and utilizing mesh information to obtain rich object features.
- Proposed a visual question answering module incorporating image captioning sentences, which successfully fuses previously generated descriptive sentences to generate high-quality answers with improved answer accuracy by using Attention+ LSTM.

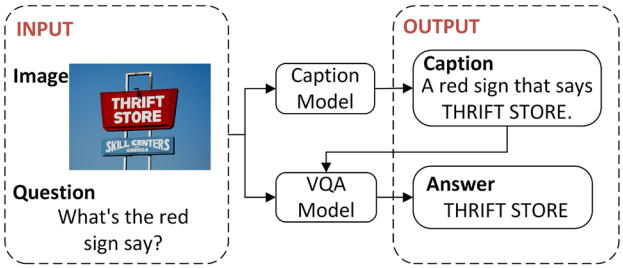

Fig. 1 Sample visual question answering based on image captioning

2 Related work

2.1 Image captioning

The image captioning task has been rapidly developed and improved through the continuous efforts of successive generations of researchers since its inception. Encoder-decoder frameworks are the basis of deep learning-based image captioning, and most of them employ CNN-RNN frameworks. Vinyals et al. [5] proposed a neural image captioning model NIC, which for the first time introduced an encoder-decoder framework into image captioning. Vaswani et al. [6] in 2017 proposed Transformer model which consists of multilayer encoder and decoder with same number of layers. Xu et al. [7] introduced the attention mechanism into image captioning for the first time and also introduced two attention-based image captioning generators. Lu et al. [8] proposed an adaptive attention mechanism with visual sentinels, which solves the problem of when and where to add an attention mechanism in order to provide meaningful information for sequence words. Anderson et al. [9] proposed a combined Top-Down and Bottom-Up attention mechanism in 2018, which is based on the Bottom-Up method of faster RCNN to obtain a set of salient image regions, each represented by a pooled convolutional feature vector. Yang et al. [10] in 2019 introduced a scene graph model in the attention mechanism, which fuses the relationship between each object, the object's own attributes, and other objects as inputs to the network by means of graph convolution. Guo et al. [11] in 2020 proposed the geometric-aware self-attention mechanism, which is an improvement of self-attention to explicitly and effectively consider the relative geometric relationships between objects in a graph, and is able to output sentences more in line with natural human language descriptions. Most of today's image captioning is about scene textual information or about specific emotions, and there is very little research on issue-specific image captioning.

2.2 Visual Question Answering

Visual question answering (VQA) [12] was first proposed in 2015 to be able to provide an accurate natural language answer given an image and a natural language question about the image. In 2016 Yang et al. [13] developed a neural network architecture called Stacked Attention Networks (SANs) for image quizzing tasks. Lu et al. [14] innovatively proposed a hierarchical problem-image synergetic attention model which is divided into three levels on the problem: words, phrases, and sentences, and

applies synergetic attention at each level, thus enhancing the model's understanding of the problem-image relationship. Ben-Younes et al. [15] proposed a multimodal tensor fusion method called MUTAN, which employs Tucker decomposition to enhance the visual question answering model's comprehension of images and text by efficiently parametrizing the bilinear interactions between the visual and text, thus improving the model performance. Kim et al. [16] proposed a bilinear attention network model BAN based on low-order bilinear pooling technique to solve the problem of attention allocation in multimodal scenarios. The BLOCK model proposed by Ben-Younes et al. [17] provides an innovative fusion model for visual question answering and visual relationship detection tasks by introducing bilinear super diagonal fusion. Gao et al. [18] proposed a model called Multi-modality Latent Interaction Network, which introduces the Multi-modality Latent Interaction potential interaction module, which is capable of learning cross-modal relationships between potential visual and linguistic representations. Lu et al. [19] proposed a multimodal reasoning approach for scientific Q&A that integrates information from different modalities by introducing multimodal data and Chain of Thought (CoT). The accuracy of visual question answering has been gradually improved, which can basically meet people's daily needs, but it still needs to be researched continuously to get better results. Wu et al. [20] present a model which jointly generates question-related captions and uses them to provide additional information to aid VQA. Li et al. [21]used Generative Adversarial Networks to generate answer embeddings conditioned on image and question inputs. At the same time, autoencoder-based techniques are combined with GAN to focus on learning optimal embeddings for questions and images to improving Visual Question Answering. Vosoughi et al. [22] modeled a confounding effect that causes language and vision bias simultaneously, then proposed a counterfactual inference to remove the influence of this effect. Models trained by this strategy can effectively reduce both visual and language biases simultaneously.

3 IC-VQA model

In this section, we introduce the main components of the IC-VQA model. As shown in Fig. 2, the IC-VQA model is mainly composed of two parts, the first part is the Caption Model, which dynamically locates the visual object region through the input question and generates image captioning about the question; the second part is the VQA Model, which generates the answer about the question by incorporating the description sentences that Improve the accuracy of answers.

3.1 Caption model

The IC-VQA model incorporates an image captioning sentence about the question in the answer generation part of the question to make the answer generation more accurate. This image captioning sentence has certain requirements, it cannot be the daily generation of captions based on the entire image content, but the combination of the image and the question, from the question to find the corresponding object in the image, so as to generate a description that is in line with the content of concern of the question, and only in this way can it have an enhancement effect on the final answer to the question. In this regard, this paper proposes the question-guided image captioning module, as shown in Fig. 3. This module consists of three main parts, the first part (orange part) is to construct the object geometric relations and use the grid information to obtain the object features; the second part (green part) is to use the attention mechanism

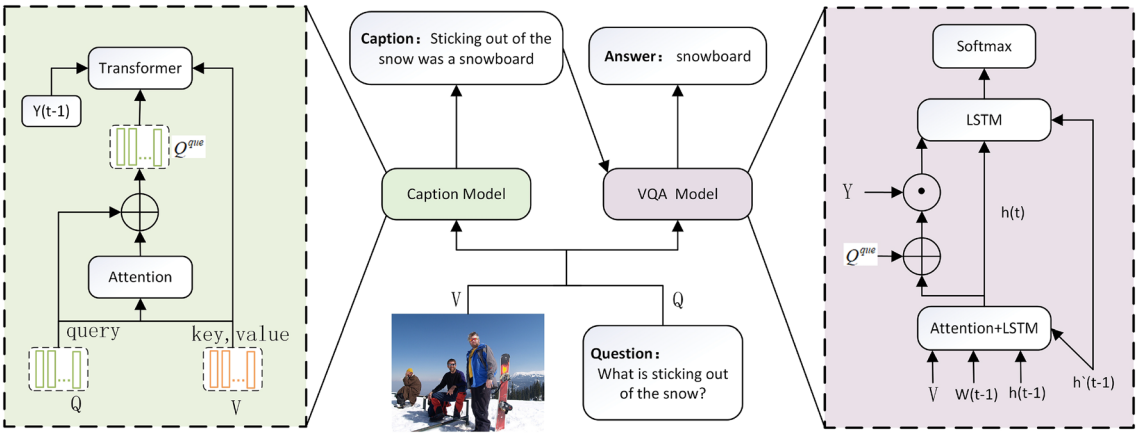

Fig. 2 Overall model diagram of IC-VQA

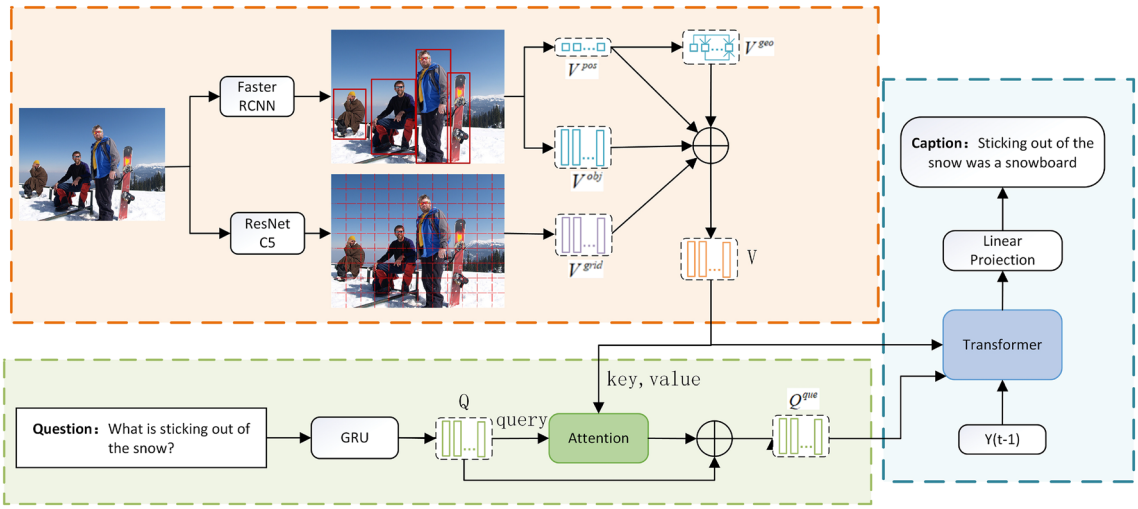

Fig. 3 Problem-guided image captioning module

to locate the object through the problem, so as to obtain the object features related to the problem and the text features of the problem; and the third part (blue part) is to output the descriptive sentences step-by-step using the Transformer. This module can successfully generate image captioning sentences for a specific problem.

1) Constructing object geometry relationships and obtaining object features using grid information.. Problems appear in a variety of forms, may be examining the count, may be examining the positional relationship of the object, etc., want to accurately locate the image position through the problem, it is necessary to establish the object geometric spatial structure, and correctly deal with the characteristics of the object. The modeling object geometry relationship in [23] is to achieve relative position encoding. Relative position information is added according to the geometric structure of the bounding boxes, and the comprehensive relation attention weight is adjusted by using relative position information. The authors model the complex visual and location relationships between input features by integrating both absolute and relative location information. We combine geometric relationships with grid features to capture both local and global information, and are able to better understand and utilize the spatial location relationship of objects in the image. The relative width and height in [24] are not calculated in the same way as ours when modeling object geometry relationship, by adding one more. According to the position between adjacent objects does not change due to viewpoint shift, the authors design geometric interaction embedding to model the position interaction of features, and use the relative position relationship between features for the calculation of learnable geometric interaction weights. We obtain the final target feature by layer normalization and summation of geometric relation features with other features. Therefore, in this paper, the image features are first extracted using Faster RCNN to obtain a set of 2048-dimensional feature vectors V^{obj} and the detection frame coordinates $V^{pos}(x_{min}, y_{max}, x_{max}, y_{min})$, V^{pos} denoting the position coordinates of the upper-left and lower-right corners of the detection frame. However, simply using the upper-left and lower-right coordinates cannot correctly determine the positional relationship between objects, so here we propose a module for constructing the geometric relationship of objects, which obtains the centroid coordinates, length and width of the object by deforming the V^{pos} coordinate, and then establishes the relative geometric relationship between any two objects, so that we can build up the spatial structure of the objects in the whole picture, and achieve the goal of locating the objects through the problem. The following is an example of the ith object:

$$(cx_i, cy_i) = \left(\frac{(x_{min})_i + (x_{max})_i}{2}, \frac{(y_{min})_i + (y_{max})_i}{2}\right) w_i$$
$$= (x_{max})_i - (x_{min})_i h_i = (y_{max})_i - (y_{min})_i$$

where (cx_i, cy_i) denotes the center point coordinates, and w_i, h_i denote the width and height of the object, respectively.

Based on[25], The relative geometric relationship between any two objects i, j is then computed from the center point coordinates, width and height of each object in three terms, where it is computed using log, which is used to measure the ratio or relative degree of some property between two objects. The formula is as follows:

$$r_{ij} = \begin{pmatrix} \log\left(\frac{|cx_i - cx_j|}{w_i}\right) \\ \log\left(\frac{|cy_i - cy_j|}{h_i}\right) \\ \log\left(\frac{w_i}{w_j}\right) \\ \log\left(\frac{h_i}{h_j}\right) \end{pmatrix}$$

$$V_{ij}^{geo} = RELU(w^T FC(r_{ij}))$$

where r_{ij} denotes the relative positional relationship between objects i, j and $r \in R^{N*N*4}$ (N denotes that there are N objects). $\log(\frac{|cx_i - cx_j|}{w_i})$ and $\log(\frac{|cy_i - cy_j|}{h_i})$ denote the horizontal distance relationship and vertical distance relationship between objects i and j, respectively. $\log(\frac{w_i}{w_j})$ and $\log(\frac{h_i}{h_j})$ denote the horizontal and vertical proportions of objects i and j, respectively. FC denotes the fully connected layer, w^T denotes the weight matrix for learning, and RELU denotes the activation function that allows for deeper modeling of the geometric relationships of the objects and extraction of more advanced feature representations. V_{ij}^{geo} represents the final learned geometric relationship between objects i and j.

After the above building object geometry relationship module, it significantly increases the connection between the objects, but it is still a bit insufficient to capture some small targets and global information, so here grid features are introduced to help the model to learn more information. Grid features help the model to detect and classify targets of different sizes and complexity, it can capture both local and global information, and it can better understand and utilize the spatial location relationship of the target in the image with high computational efficiency.

Here grid features are extracted from the image using the C5 hierarchy in the ResNet [26] network, which are a tensor F of shape (H,W,C), where H and W represent the height and width, and C represents the number of feature channels. Then, the mesh features are reshaped into the form of (m,2048) using the AdaptiveAvgPool2d function, where m represents the size of the spatial dimension of the output features, i.e., the height and width of the output feature map. The specific operation is to use AdaptiveAvgPool2d(m,m) to perform an adaptive average pooling operation on the input mesh features to generate a feature map of size (m,m,C), where C denotes the number of feature channels at each position. Then, the dimension of the feature map is transformed to change the (m,m,C)-sized feature map into the form of (m,2048), i.e., the (m,C)-sized feature vector at each position is mapped to a feature vector of length 2048. After the above operation the grid features extracted from the image can be reshaped into a feature representation with fixed dimensions for subsequent processing and analysis, which can reduce the dimensionality of the features while retaining the important information and make the feature vectors more operable and consistent. The formula is as follows:

$$F_{pool} = AdaptiveAvgPool2d(F, m)$$

$$V^{grid} = Reshape(F_{pool}, (m^2, C)) \times Linear(C, 2048)$$

Where F denotes the mesh feature vector extracted from the ResNet network, F_{pool} denotes the feature vector after adaptive average pooling, V^{grid} denotes the feature vector after dimensionality change, Reshape denotes the reshape operation, and Linear denotes the linear change operation.

The object features V^{obj}, detection frame position features V^{pos}, object geometric relationship features V^{geo}, and mesh features V^{grid} are now obtained, and all the features are layer normalized and summed to obtain the final object features V.

$$V = LN(w_1 V^{obj}) + LN(w_2 V^{pos}) + LN(w_3 V^{geo}) + LN(w_4 V^{grid})$$

where LN denotes layer normalization and w_1, w_2, w_3, w_4 denotes weighting factor.

2) Utilizing Attention Mechanisms to Locate Objects through Questions. To generate image captioning about a problem, it is necessary to localize different visual regions for different problems and dynamically select corresponding visual objects. As shown in Fig. 3, the question is what is sticking out from the snow, so it should be localized to skis, and if it is localized to a person, the captioning result must be wrong. Therefore, it is crucial to localize the corresponding visual object for each problem before generating the description. Using the attention mechanism to take the problem feature $Q = [q_1, q_2, ..., q_n]$ as query and the visual feature $V = [v_1, v_2, ..., v_n]$ obtained earlier as key and value, the computation steps are as follows. A special recurrent neural network GRU is used here to encode the questions.

$$s_{ij} = q_i W_q (v_j W_k)^T$$

$$a_i = Softmax([s_{i1}, s_{i2}, ..., s_{in}])$$

$$V_i^Q = a_i V$$

where W_q, W_k denotes the learnable matrix, s_{ij} denotes the attention value of the jth visual feature corresponding to the ith token in problem Q, a_i denotes the visual attention weight distribution of the ith token in the problem, and V_i^Q denotes the attended visual feature.

To generate descriptive sentences that accurately answer the question, it is not enough to simply know the visual features being attended to. For example, the question is What color is the puppy? Is the puppy happy? What is the puppy playing with? All three questions localize to the visual object which is a dog. If we only know the visual features but not the textual features of the questions, i.e., we do not know what the questions are asking, we cannot generate the correct answers. Therefore, it is important to find not only the visual features

that the questions focus on, but also to know the textual features of the questions, and the sum of the two is calculated as follows:

$$Q_i^{que} = V_i^Q W_v + q_i W_q'$$

Where W_v, W_q' denotes the coefficient matrix and Q_i^{que} denotes the final features and feeds them into the Transformer decoder.

3) Step-by-step output of descriptive sentences using Transformer. Finally, the output is decoded step-by-step using the Transformer decoder, and at each time step t, the input to the Transformer layer has three parts: the visual feature V, the textual features of the problem and the visual features of interest Q^{que}, and the output from the previous step y_{t-1}, which is then passed through a linear layer to calculate the current score and output it.

$$y_t = W z_{t-1}^{trans} + b$$

where W and b denote learnable parameters, z_{t-1}^{trans} denotes the output of the previous step after the Transformer layer, and y_t denotes the step-by-step output.

3.2 VQA model

From section 3.1 four parts have been obtained: the visual object features of the image $V = [v_1, v_2, ..., v_n]$, the question features $Q = [q_1, q_2, ..., q_n]$, the textual features of the question and the visual features of the concern Q^{que}, and the descriptive sentence $Y = [y_1, y_2, ..., y_n]$. Taking time step t as an example, the Q&A output module is shown in Fig. 4. The main framework used is Attention+LSTM, and the LSTM unit consists of input gates, output gates, and forgetting gates, and the inputs to each gate are the visual feature V, the current question word q_{t-1}, the hidden state of the previous step h_{t-1}, and the answer word generated in the previous step w_{t-1}.

Both image captioning and VQA need to output the results step by step, adjusting the region of interest of the image according to the output of the previous step, so as to accurately predict the next word. Here firstly Attention+LSTM is used, the inputs are visual feature V, the output word w_{t-1} from the previous step, and the hidden states h_{t-1} and $h_{t-1}^{'}$ of the two LSTM, which makes the model to find the region of the image that should be focused on at this time, and the operation is as follows:

$$h_t = LSTM_{Attention}(V, w_{t-1}, h_{t-1}, h_{t-1}^{'})$$

Then, the features h_t obtained in this step are weighted and fused with the problem text features and problem concern visual features Q^{que} obtained in section 3.1 to obtain the

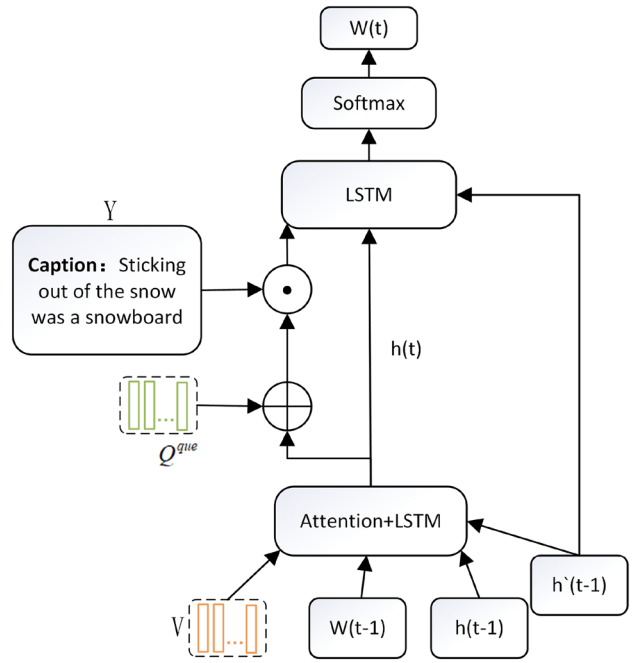

Fig. 4 VQA output module

overall features v_{he}. The generated descriptive sentence Y is multiplied element-by-element with the overall features v_{he}, which fuses the feature information of different modalities, so that it can learn richer and more comprehensive feature representations, which helps the model to better understand the relationship between the image and the text connection between image and text.

$$v_{he} = Q^{que} W_1 + h_t W_2$$

$$v_{fin} = Y v_{he}$$

where W_1, W_2 denotes the learnable matrix.

The obtained final features v_{fin} are passed into the second LSTM to compute the predicted words.

$$h_t^{'} = LSTM(v_{fin}, h_t, h_{t-1}^{'})$$

In this paper, the IC-VQA model is capable of generating answers for multiple words, so the VQA is trained to maximize the probability of the answer to a given image and question, so that the answers generated by the model are as close as possible to the probability of the actual answer.

$$w_t = softmax(w_3 h_t^{'} + b_3)$$

$$\log probability(W/I, Q) = \sum_{t=1}^{n} \log probability(w_k / w_{1:k-1}, I, Q)$$

where $probability(w_k/w_{1:k-1}, I, Q)$ denotes the probability of the answer word w_k, $w_{1:k-1}$ denotes the word generated in the first k-1 steps, I denotes the image, and Q denotes the question.

4 Experiments

4.1 Datasets and evaluation metric

There are three datasets used in this paper, MSCOCO [12], VQA1.0 [27] and VQA2.0 [27], as shown in Table 1. MSCOCO is used for image captioning, and VQA1.0 and VQA2.0 are used for visual question answering.Where VQA1.0 and VQA2.0 contain images from MSCOCO dataset which can be corresponded by image_id.

The MSCOCO dataset was constructed by Microsoft in 2014 with a lot of labor and resources, and there are 118,000, 5,000, and 40,775 images in the training, validation, and test sets, respectively. The image content is mostly natural scene images or daily life images with complex backgrounds, and each image has five different descriptive pairs, which is the most popular and widely used dataset in the field of computer vision nowadays.

VQA1.0 is the earliest VQA dataset, with a training set of about 82,783 images and a validation set of about 40,504 images, with about three questions per image and about ten answers per question, where one standard answer is specified. The test set has about 81,434 images, again with three questions per image, but no test set answers are provided for model prediction and evaluation on unknown answers. VQA2.0 improves and extends VQA1.0 with more varied and complex questions and answers, and more images and question-answer pairs to make the dataset more rich, comprehensive, and challenging. VQA2.0 training The VQA2.0 training set consists of about 200,000 images containing about 443,000 question-answer pairs; the validation set consists of about 100,000 images containing about 214,000 question-answer pairs. The test set has about 200,000 images containing over 447,000 question-answer pairs, and again, no answers to the questions are provided. The answers in both the VQA1.0 and VQA2.0 datasets contain a variety of types: the binary (yes/no) type Yes/No, the numeric type Number, and the other type Other, and the experiments in the following section evaluate the answers using these three types and the combined type Overall for evaluation.

The most commonly used evaluation metric for VQA is the accuracy rate, which is calculated by comparing the answers generated by the model with the answers labeled by humans to measure the accuracy of the model in answering the questions. Currently, the highest level that humans can achieve is 80.83%, and in this paper, we use the accuracy rate to evaluate against the four types of Yes/No, Number, Other and Overall. The formula for calculating the accuracy rate is as follows:

$$Acc(w_i) = min\left(\frac{count(w_i)}{3}, 1\right)$$

where $count(w_i)$ denotes the number of annotators who have the same answer w_i as the model, and $Acc(w_i)$ denotes the w_i final accuracy obtained; if the model agrees with the answers of three or more annotators, it will get a perfect accuracy for a problem, a metric that can be very helpful for ambiguity problems.

4.2 Experiments results and analysis

1) Experiments setup. The visual object embedding vector size was 2048, the maximum length of the generated descriptive sentence was set to 30, the hidden layer dimension of the LSTM, the visual features,and the vectors representing the word embeddings were converted to 512, and the number of heads of the Attention in the VQA module was 8. The maximum number of words in the problem was set to 14, and each word was converted to lower case. The model was trained using Adam optimizer. The model was trained for 12000 iterations.The learning rate was set to 0.0005 with a decrement of 0.6 times every 4 periods and the batch size size was 64.

2) Comparative Experiments. In this paper, two datasets, VQA1.0 and VQA2.0, are used to conduct experiments, and some more typical models are selected for comparative analysis to judge the model performance, and the results are

Table 1 Comparison of datasets

Dataset name	Number of images/sheet			Number of descriptions per image	Number of questions per image	Range of application
	Training set	Validation set	Test set			
MSCOCO	118000	5000	40775	5	–	Image captioning
VQA1.0	82783	40504	81434	–	3	Visual question answering
VQA2.0	200000	100000	200000	–	≥ 3	

Table 2 Comparison results on the VQA1.0 dataset

Method	Accuracy			
	Number	Yes/No	Other	Overall
Ibowimg [28]	34.98	76.76	42.62	55.89
SAN [29]	37.53	80.8	43.48	58.24
RAU [30]	38.2	81.7	52.8	63.2
DAN [31]	38.1	82.8	54	64.2
String Baseline [32]	39.1	82	55.2	64.6
MLAN [33]	40.9	83.7	53.7	64.8
MLB [34]	37.9	84.02	54.77	65.07
DCN [35]	41.27	84.61	56.83	66.66
VRR [36]	41.33	84.18	58.58	67.33
CAM [37]	42.36	**86.55**	**60.38**	69.29
IC-VQA(Ours)	**42.53**	84.12	57.81	**70.1**

Values highlighted in bold indicate the best-performing metric for the respective type

Table 3 Comparison results on the VQA2.0 dataset

Method	Accuracy			
	Number	Yes/No	Other	Overall
VQA team-Prior [38]	0.36	61.2	1.17	25.98
VQA team-language onl [38]	31.55	67.01	27.37	44.26
VQA team-LSTM+CNN [38]	35.18	73.46	41.83	54.22
MCB [38, 39]	38.28	78.82	53.36	62.27
Adelaid [40]	39.77	79.32	52.59	62.27
BLIP-2 ViT-g FlanT5XX [41]	–	–	–	65.0
Adelaide+Detecto [40]	43.9	82.2	56.26	65.67
DC [35]	46.93	83.89	56.9	67
AT-CM [42]	47.72	85.85	60.35	69.55
Caption VQA [20]	47.41	86.15	60.41	69.66
UP-down [9]	**48.64**	86.60	61.15	70.34
JE-MHA [43]	–	–	–	70.79
IC-VQA(Ours)	46.83	**87.2**	**61.98**	**70.85**

Values highlighted in bold indicate the best-performing metric for the respective type

shown in Table2 and Table 3 respectively. The evaluation index is the accuracy rate, which is analyzed for the four types of Yes/No, Number, Other and Overall, where Overall indicates the overall type, which is the most representative. The highest values of the results for each type are bolded for ease of observation.

Observing the results in the table, it can be found that the IC-VQA model in this paper performs well on both datasets. The accuracy of the IC-VQA model on Number and Overall types in the VQA1.0 dataset reaches the maximum value, and the accuracy of Yes/No type is the second largest value. The accuracy of the Other type is at the upstream level in the VQA1.0. The accuracy of the IC-VQA model on the Yes/No, Other and Overall types in the VQA2.0 dataset reaches the maximum value, and the accuracy of Number type is the third largest value. The accuracy on the overall metric Overall on both datasets reaches 70.1 and 70.85 respectively, which are the highest values, which proves the effectiveness of the IC-VQA model to successfully improve the accuracy of the visual question answering results.

3) Ablation experiment. The IC-VQA model as a whole can be divided into two parts, the first part is to generate the image captioning about the question (Caption Model), and the second part is to incorporate the image captioning to output the question answer (VQA Model). There are two key steps in the first part which are: obtaining rich object features (denoted as Obj) by constructing object geometric relations and utilizing mesh information, and finding the visual object of interest for the question (denoted as Att-obj) by using the attention mechanism. There are also two key steps in the second part, which are: incorporating image captioning sentences (denoted as Cap), and using the attention mechanism to predict the object region for the next word (denoted as Att-w). Therefore, ablation experiments are conducted here for these four key steps to verify the effectiveness of each of their parts. Ablation experiments are first conducted on the dataset VQA1.0, and the results are shown in Table 4. Row 1 in the table indicates that without using these four modules, i.e., without incorporating the image captioning sentence and without using the attention mechanism, the question and image are directly inputted into the LSTM framework to output the results, and the overall accuracy (Overall) of the results is only 52.45, while the results of row 2, which

Table 4 Results of ablation experiments on VQA1.0 dataset

	Obj	Att-obj	Cap	Att-w	Accuracy			
					Number	Yes/No	Other	Overall
1					34.23	73.4	50.7	52.45
2				✓	40.38	79.16	53.78	63.5
3			✓		38.4	77.25	52.19	56.73
4		✓	✓		40.1	79.32	55.98	66.46
5	✓	✓	✓		41.26	80.42	56.12	67.81
6	✓	✓	✓	✓	**42.53**	**82.3**	**57.81**	**70.1**

Values highlighted in bold indicate the best-performing metric for the respective type

incorporates the attention mechanism in the visual question answering part, directly increase by 11 points to 63.5. Comparing rows 5 and 6, the accuracy of Overall increases by 3 points after incorporating the Att-w. mechanism is added to the visual question answering part in line 2 and the result directly increases by 11 points with an accuracy of 63.5. Comparing lines 5 and 6, the addition of Att-w improves the Overall accuracy by 3 points. Both parts demonstrate the importance of adding attention to the VQA Model module. Looking at row 3 in the table one can see that the results go up by 4 points when only descriptive sentences are added to the model. Compare row 4 which uses the attention mechanism when generating the descriptive sentence, making the descriptive sentence relevant to the problem, and the results go up by 14 points. Compare row 5, which not only uses the attention mechanism but also adds rich visual features when generating descriptive sentences, so the result reaches 67.81, which is 15 points higher than that of row 1, 11 points higher than that of simply adding descriptive sentences, and 1 point higher than that of generating descriptive sentences using only the attention mechanism. The reason for this analysis is that when adding image captioning sentences alone, the sentences generated are sometimes relevant and sometimes irrelevant to the question, so the results are appropriately enhanced. Line 4 generates sentences that are highly relevant to the question, so there is a significant rise in accuracy. Line 5 adds rich visual features to line 4 to find more accurate objects based on the problem, and generates descriptive sentences with the strongest relevance to the problem. Therefore, it proves the effectiveness of the Caption Model module, which can accurately generate descriptive sentences about the question; it proves the effectiveness of the Caption Model module, which incorporates rich visual features; and it proves the effectiveness of the VQA Model module, which incorporates descriptions about the question to improve the accuracy of answering the question.

Next, ablation experiments are performed on the dataset VQA2.0 and the results are shown in Table 5. Observing the results, it can be found that the overall accuracy (Overall) of the model in row 6 is 70.85, which is higher than all the other rows. So, the same proves the validity of the modules in this paper's model IC-VQA and proves the validity of the IC-VQA model.

4) Visualization and analysis. In order to more intuitively observe the performance of the model IC-VQA on the dataset, some representative images are selected for visualization and analysis, and the experimental results are shown in Fig. 5. Images A and B are from the dataset VQA1.0, and image C is from the dataset VQA2.0. VQA1.0 asks three questions for each image, and VQA2.0 makes additional corrections to VQA1.0, and the number of questions is generally greater than or equal to three, with an indeterminate number of specifics, e.g., the selected image C contains four questions.

Observing the experimental results, it can be found that for each question IC-VQA model can generate a relevant description and the description sentence can enhance the Q&A output results. For some simple questions, such as Question 3 in Figure A: How many windows are there? the model can successfully locate the object of windows and output a result of windows with an answer of type NUMBER. Another example is Question 1 in Figure C: Is the color of a man's tie purple? The description module is able to output the color of the tie as green, and the Q&A module successfully outputs the answer as no, which is of type yes/no.

For some more complex problems, such as problem 2 in Figure B: Does the food have fluff? In this paper, the description model cannot directly determine whether this food has fuzz or not, but it can determine that the food is apples and oranges, and generate the relevant descriptions, because apples and oranges are smooth on the surface, so it can also enhance the final result. For example, Question 4 in Figure C: Is the tie a name brand? For this question in this paper the description model can't make a direct judgment, it can only locate that the object is a tie TIE and then output a description about the tie. So the answer to this complex question is the shortcoming of this paper's model, which can be used as a subsequent research direction.

To summarize, the IC-VQA model has good experimental results and can successfully generate image captioning sentences for specific questions, the description sentences have

Table 5 Results of ablation experiments on VQA2.0 dataset

	Obj	Att-obj	Cap	Att-w	Accuracy			
					Number	Yes/No	Other	Overall
1					38.46	79.32	51.56	62.8
2				✓	41.69	83.47	59.4	68.87
3			✓		41.72	83.68	57.3	64.72
4		✓	✓		43.24	85.02	59.12	69.2
5	✓	✓	✓		44.63	85.73	60.23	70.13
6	✓	✓	✓	✓	**46.83**	**87.2**	**61.98**	**70.85**

Values highlighted in bold indicate the best-performing metric for the respective type

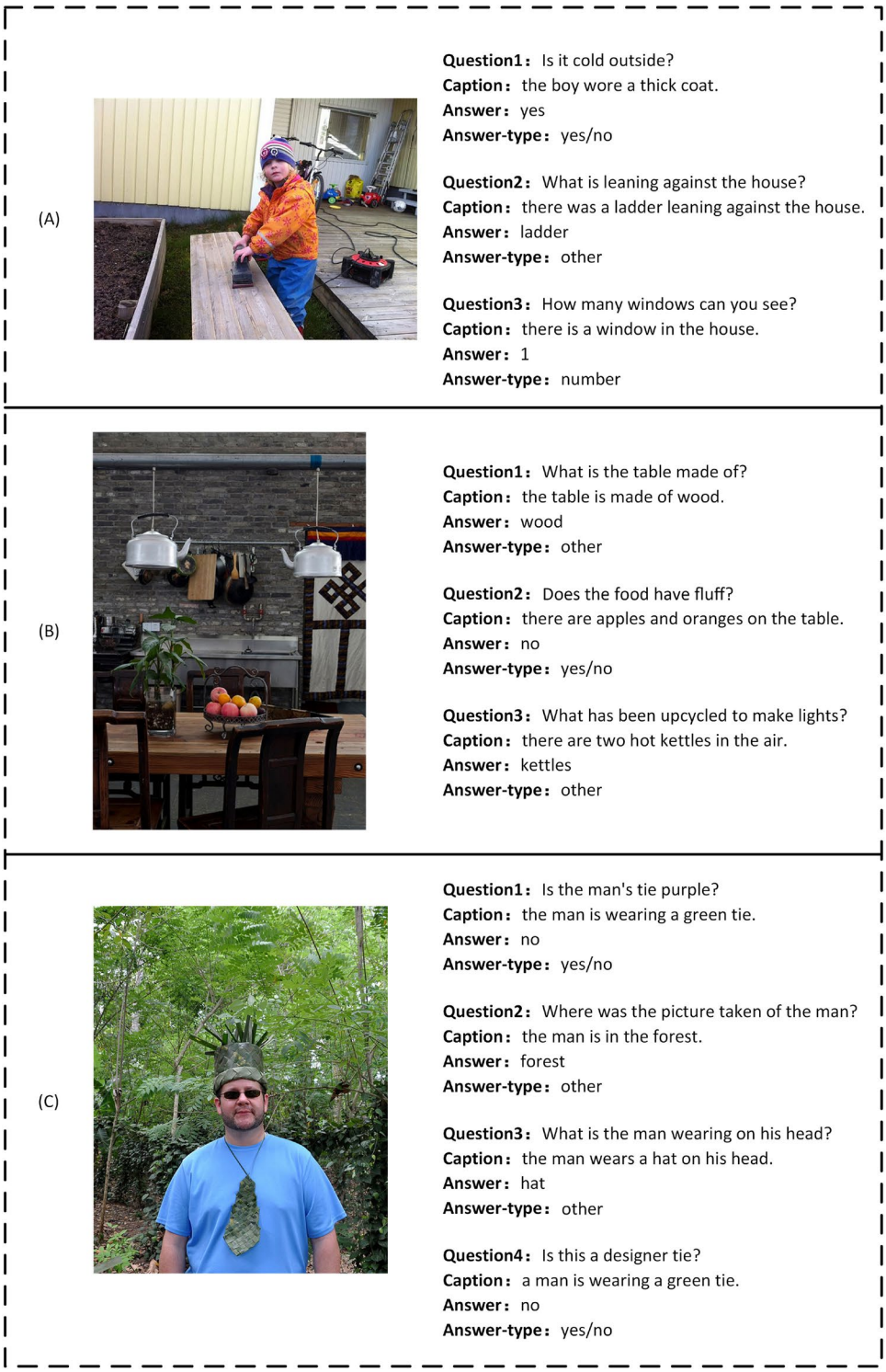

Fig. 5 Visualization of VQA1.0 and VQA2.0 experimental results

a certain enhancement effect on the results of visual question answering and can improve the accuracy of the answers, and the results can meet the expectations.

Because we want to establish the connection between the two task datasets of image captioning and visual question answering, we can only study the direction without scene text information, which leads to the IC-VQA model does not include the treatment of scene text information, so the model effect is not very good for those questions with text information. However, the error case is shown in Fig. 6. Problem 2 is what is the color of the Salisbury Rd sign, the IC-VQA model made an error in recognizing the

Question1: How many signs?
Caption: there are four signs.
Correct Answer: 4
Our Answer: 4
Answer-type: number

Question2: What color is the Salisbury Rd. sign?
Caption: a white sign that says salisburyr.
Correct Answer: white
Our Answer: white
Answer-type: other

Question3: What does the last sign say?
Caption: there are white and yellow signs.
Correct Answer: ross st
Our Answer: slros
Answer-type: other

Fig. 6 Visualization of VQA1.0 and VQA2.0 experimental results

word Salisbury Rd, and the recognition result is salisburyr, so the generated description sentence is wrong. Question 3 is what is written above the last sign? In this paper, the description result outputs a completely irrelevant sentence, and the Q&A result is also a completely wrong word. The reason for analyzing the error is that the model doesn't know which is the last sign, can't recognize the text information correctly, and can't build the relationship between text and object correctly. So these problems can be the next research direction.

5 Conclusions

In this paper, we propose a visual question answering model IC-VQA based on image captioning, which can generate an image description sentence about the question based on the input image and the question, and then incorporate this sentence into the visual question answering module, significantly improving the accuracy of the generated answer.The IC-VQA model is used on the Comparison experiments, ablation experiments and visualization of the results on the datasets VQA1.0 and VQA2.0 are performed to analyze the correct and incorrect cases, and the results show the effectiveness of the IC-VQA model in narrowing the gap with humans. Because we want to establish the link between the two tasks of image description and visual quizzing at present, we can only do it through the MSCOCO dataset, so the research is in the direction of the direction without scene text information, which leads to the final model for the detection of scene text recognition is not high, or even error. In the future, we will try to establish the linkage of datasets about scene text for both tasks of image description and visual quizzing, so that the model will be more effective.

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Author contributions Kun Zhou and Qiongjie Liu wrote the entire paper. Dexin Zhao reviewed the paper and suggested important revisions.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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